

Motor Imagery EEG Signal Classification for Stroke Survivors Rehabilitation

Alex Efstathios Voinas¹, Rig Das², Muhammad Ahmed Khan², Iris Brunner³ and Sadasivan Puthusserypady²

^{1,2}Department of Health Technology, Technical University of Denmark, Kgs. Lyngby 2800, Denmark.

³Hammel Neurocenter, Aarhus University, Voldbyvej 15, 8450 Hammel, Denmark.

¹s203021@student.dtu.dk, ²{rigdas, mahkh, sapu}@dtu.dk, ³Iris.Brunner@rm.dk.

Abstract—Motor Imagery (MI) based Brain Computer Interface (BCI) is a promising neurorehabilitation tool for treating motor impaired stroke survivors. It enables the MI electroencephalogram (EEG) signals to be converted/mapped into customized robotic and assisting commands. Even though stroke causes varying effects in the brain, the EEG signals have shown promises towards the MI classification of post-stroke subjects. This paper presents a MI based left and right wrist dorsiflexion classification performed on 6 stroke subjects. The MI EEG data are recorded from 16 electrode locations on the subject's scalp. Three different feature extraction methods are used to compare and find the best performing one; the first one is based on the Wavelet Packet Decomposition (WPD) combined with Higher Order Statistics (HOS) which is compared with the widely used Common Spatial Pattern (CSP) and Filter Bank Common Spatial Pattern (FBCSP) filter method. The MI classifications are performed using Random Forest (RF) algorithm to achieve a mean accuracy that exceeds 70% for the WPD+HOS method, while outperforming the CSP and FBCSP based methods.

Index Terms—Brain Computer Interface (BCI), Motor Imagery (MI), Stroke-Rehabilitation, Wavelet Packet Decomposition (WPD), Random Forest (RF).

I. INTRODUCTION

Stroke is caused by blood flow disruptions to the brain. Through the modern human lifestyle, stroke has turned out to be one of the main causes of long-term disability globally, leaving a big majority of its survivors with motor impairments and limb paralysis [1]. Brain Computer-Interface (BCI) is shown to be one of the effective methods for motor neurorehabilitation of post-stroke impaired patients as it makes possible the decoding and translation of brain electrophysiological signals into a computer input [2]–[4].

Motor Imagery (MI)-based BCI involves MI tasks, the mental process of imagining the movement of a body part, such as a limb, without actually moving it or activating any related peripheral muscles [5]. It is the leading non-invasive method for treating arm disabilities [6]. The working principle of MI-BCI systems is that the subjects have to focus on imagining the MI task while the electroencephalograms (EEG) are recorded from their brain to capture the spatio-temporal variations caused by the activation of the specific cortex areas associated with the imagined movements of a body part [7]. The main area of interest is the motor cortex [8]. The manual extraction of MI features from EEG comes

with obstacles, as it requires the subject's full attention and willingness while imagining the movements and also there are artifacts corrupting the signal (non-cerebral signals, such as the eye-blinking, power-line interference, etc.) that needs to be overcome [9]. Therefore the aim is to automate this process with the use of machine learning (ML) and other statistical methods, i.e., by training a classifier to distinguish between the different imaginary tasks, in order to translate it into computer commands. Offline MI-BCI classification studies, where the EEG recordings and their analysis take place at different times, generate useful insight and pave ways for fully autonomous online post-stroke limb rehabilitation systems [4].

In this paper, we explore an offline MI-signal classification pipeline for separating the imaginary left and right wrist dorsiflexion EEG signals. The EEG signals are collected from 6 real stroke survivors and each of these subjects is having varying levels of paralysis or disabilities. The EEG signals are acquired using a 16-channel RecoveriX [10] system and then pre-processed to increase the signal-to-noise ratio. The pre-processed data are then used for feature extraction using both the time-frequency and spatio-temporal analysis. In the first case, we use Wavelet Packet Decomposition (WPD) technique, where the signals are decomposed into frequency sub-bands on which the Discrete Wavelet Transform (DWT) is applied to find a sequence of coefficients. Then a Higher Order Statistics (HOS) is applied to extract the features. In the second case, i.e., the spatio-temporal, the Common Spatial Pattern (CSP) filter and Filter Bank Common Spatial Pattern (FBCSP) filter are used for feature extraction. For all the cases, the Random Forest (RF) algorithm is used as a classifier.

This paper is organized as follows: Section II discusses the state-of-the-art techniques that are currently in use for stroke survivors' and healthy controls' MI-EEG signal classification. Section III talks about the employed method for feature extraction and classification of MI-EEG signal and section IV shows the performance achieved by our proposed method. Finally the conclusions are drawn in section V.

II. STATE-OF-THE-ART REVIEW FOR MI EEG SIGNAL CLASSIFICATION OF STROKE SURVIVORS

There exists a number of state-of-the-art studies in the field of MI-task evoked EEG signal classification for healthy control subjects and stroke survivors. Most of these studies are

TABLE I
STATE-OF-THE-ART METHODS OVERVIEW FOR MI-EEG SIGNAL CLASSIFICATION

	Paper	Year	#Subjects	#Channels	Features	Classification	Accuracy
Stoke Survivors	[11]	2014	10	45	CSP	LDA	0.9 (AU-ROC)
	[12]	2016	5	16	CSSBP	SVM	70%
	[13]	2018	5	64	CSP	LDA	87.4%
	[14]	2020	16	64	PLV	Naive Bayes	68.63%
	[15]	2021	10	12	SW-LCR/SW-Mode, CSP	LDA	80%
Healthy Controls	[16]	2019	9	22	CSP, WPD	SVM	93.75% (pooled CV)
	[17]	2020	9	22	CSP	SVM+KNN+LDA	83%
	[18]	2021	1	3	CSP+DWT, CSP+WPD	SVM, SRC-EN/SRC-L1	91%

performed on upper limbs' movement related tasks as there are enough publicly available datasets. These state-of-the-art methods serve as a benchmark for comparing all the newly invented methods' performances. Most of these studies mainly focus on a large number of EEG recording channels to improve their performance accuracy. Some of these methods for MI-EEG signals' classification are revisited in this section. Table I shows a detailed comparison of these methods separated by stroke survivors and healthy control subjects. The following two sub-sections discuss about the existing state-of-the-art methods used for stroke survivors and healthy controls' EEG signal classification.

A. Stroke Subjects' Studies

Tangwiriyaakul et al. in [11], conducted a study on 10 hemispheric acute stroke survivors to classify imaginary hand movements. Linear Discriminant Analysis (LDA) is used to classify the features that are extracted using CSP. The EEG recording is initially performed using 3-11 electrodes that are spread over the motor cortex. Later, the authors have shown that the accuracy of classification can be improved by increasing the number of channels to 45, which are spread beyond the motor cortex area. An average area under the receiving operating characteristic (AU-ROC) of 0.7 and 0.9 are achieved while using and not using CSP features, respectively. Therefore, it can be seen that the application of CSP has no significant effect in improving the performance. Liu et al. [12] proposed an adaptive boosting algorithm, termed the Common Spatial-Spectral Boosting Pattern (CSSBP), for overcoming the performance obstacles of CSP. A 16-channel EEG recording system is used to acquire EEG data from 5 stroke survivors and a maximum mean classification accuracy of 70% is achieved. In [13], authors have used RecoveriX system for collecting 64 channel EEG data from imaginary wrist dorsiflexion of 5 stroke survivors. CSP is used for feature extraction and LDA for classification. An average accuracy of 87.4% is achieved for their proposed method, which is one of the best performance achieved in the case of stroke survivors. They have used functional electrical stimulation (FES) to move the subjects' hand, whenever a MI task is correctly classified. They showcased that the stroke survivors can control a MI-BCI system with a relatively high accuracy. They also hinted that the stroke survivors are highly motivated to participate in such studies in order to improve their motor abilities. Benzy et

al. [14] deviated from the left/right imaginary hand movement classification and performed a study on predicting the intended hand movement direction. The EEG data is collected from 16 subjects using a 64-channel EEG system. The classified EEG data are used to provide control commands to a motorized arm support. The synchronization measure called phase locking value (PLV), extracted from EEG, is used as feature for decoding the imaginary directional movements. The event-related desynchronization/synchronization (ERD/ERS) analysis was also performed. Binary classification is performed using a Naive Bayes classifier to achieve a classification accuracy of 68.63%. In [15], Gaur et al. conducted a study on imaginary hand movement classification of 10 stroke subjects from [19] and validated on the BCI Competition IV dataset 2a (9 healthy subjects). A 12-channel EEG system is used for data collection. They proposed an improved method for handling the intersession non-stationarity effects involving two sliding window-based techniques. One calculating the longest consecutive repetition and the other calculating the mode of the prediction sequence of all the sliding windows, termed as SW-LCR and SW-Mode, respectively. The pair of CSP and LDA was used for extracting features and classification to achieve an average accuracy of 80%.

B. Healthy Subjects' Studies

Regarding the state-of-the-art methods performed on healthy controls' data, below are three recent methods that used wavelet frequency related methods for MI-EEG signal classification. Feng et al. in [16] proposed a method that combined CSP and 3rd order WPD to classify MI EEG signals. BCI competition datasets are used by the authors to test their proposed method while using support vector machine (SVM) as classifier. Training and testing is performed on all the subjects to achieve a classification accuracy of 93.75%. In [17], Fan et al. used BCI competition IV Dataset-2a (9 healthy subjects, 22-channel EEG) and proposed an ensemble classification method by combining the SVM, KNN (k-nearest neighbor) and LDA classifiers with a final voting system. The ensemble classifier is used on top of CSP extracted features. Authors have showcased that the ensemble method achieved an increase in average accuracy of 4% in comparison with classification which uses only SVM as a classifier. Deng et al. in [18] performed a left/right hand MI study based on the BCI competition dataset II, and III, where 3-channel EEG

data of a single healthy subject is used. They investigated the efficacy of combining spatio-temporal and frequency features. Different methods, such as the CSP, DWT and WPD are used for feature extraction. SVM, and sparse representation classification (SRC) with elastomeric network (EN) and L1 Norm are the classifiers used by the authors. They demonstrated that fusing different feature types can have a positive effect on the predicted accuracy. A maximum accuracy of around 91% is achieved using the combination CSP+WPD/CSP+DWT with SVM.

III. PROPOSED METHODOLOGY

Three different sets of experiments, using WPD, CSP and FBCSP as feature extraction method, respectively are performed here to find the best suited method for MI-EEG signal classification. For the classification, only RF is used for all the cases as there are binary set of tasks (left and right wrist dorsiflexion) to classify. A set of pre-processing steps are performed before the feature extraction to increase the signal-to-noise ratio. The pre-processing, feature extraction and classification steps are elaborated in the following subsections and the flow diagram of the proposed method is depicted in Fig.1.



Fig. 1. MI-EEG data classification flow diagram.

A. Dataset and MI-Tasks

MI-EEG data, for this experiment, are collected at Hammel Neurorehabilitation and Research Centre, Aarhus University, Denmark. Data are collected using the RecoveriX system of g.tec [10]. A total of 6 stroke survivors' data have been collected in 71 different recording sessions of 20-30 minute duration (each). Subjects are asked to imagine left or right wrist dorsiflexion. The timing information of a single such task of dorsiflexion is illustrated in Fig.2. At the beginning of a task, the subjects hear an acoustic signal that announced randomly left or right hand imagination. The subjects look at avatar hands on the screen which lifted if the subjects succeeded to imagine the movements appropriately. This imagining of wrist dorsiflexion continued for another 6 seconds, then the subjects are prompted to relax. The EEG signals are collected from 16 channels, i.e. FC_5 , C_5 , CP_5 , C_3 , FC_1 , C_1 , CP_1 , FC_Z , C_Z , FC_2 , C_2 , CP_2 , C_4 , FC_6 , C_6 , CP_6 . The sampling frequency of the recorded data is 256Hz. A summary of the number of events per subject is provided in Table II.

B. Pre-processing

In the pre-processing step, to remove the artefacts and increase the SNR [4], the EEG data sessions are band-pass (BP) filtered in the range 4-40Hz using forward-backward zero phase filtering with an FIR filter of order $n = 75$. Choosing the

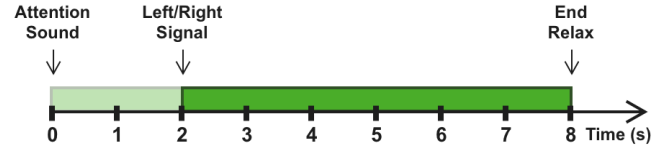


Fig. 2. Single MI task (event): in dark green the duration used in analysis

TABLE II
OVERVIEW OF THE NUMBER OF EVENTS PER SUBJECT

	S1	S2	S3	S4	S5	S6
Right Wrist	497	752	950	560	990	1069
Left Wrist	485	753	948	560	990	1066

filter order is the result of finding the optimum between low computational effort (low order) and high frequency resolution (high order), and the order of $n = 75$ gave the best results, meeting both the requirements equally [9]. After the BP filtering, events are isolated and only the data between the (2–8) second, i.e. 6 second data, are extracted as illustrated in dark green color in Fig.2. Then a common average referencing (CAR) filter and a data cleaning steps are applied, where the events with the ratio of their maximum to minimum signal difference being greater than 4 (per channel) are dropped. The output of this step is a data of 4384 events, reduced by 20% after the cleaning stage.

C. Feature Extraction using WPD and HOS

WPD is ideal for analyzing non-stationary signals like EEG as it combines both time and frequency domain information [20]. WPD is an extension to the standard DWT and works by decomposing a discrete-time signal into sub-bands (wavelet coefficients) by scaling and shifting a chosen mother wavelet (MW) [21]. The final number of sub-bands are determined by the chosen WPD order. Here, a 4th order WPD is used to decompose each signal into $2^4 = 16$ frequency bands, using as MW the Daubechies 2 (db2) wavelet. The frequency band for extracting the wavelet packets, is kept as full band-pass of range 4-40Hz. Hence providing the minimum frequency resolution of $f = 2.25\text{Hz}$ after the decomposition [16]. For each of these sub-band signals, the *Mean* and *Average Power* of their coefficients, and the HOS of *Skewness* and *Kurtosis* are extracted [22]. This process is repeated for all the 16 channels and are stacked together into a long 1-dimensional vector (one per event). This constitutes the features for the classifier.

D. Feature Extraction using CSP

CSP method is one of the widely adopted ones for EEG signal's feature extraction as it is able to produce highly accurate results on the separation of two class (binary) MI tasks. Its principle of operation is that it maximizes the variance in one class, while minimizing the variance of the other class [23], [24]. The CSP-filtered EEG signal is denoted as $\mathbf{X}_{CSP} \in \mathbb{R}^{n_{Chann} \times T}$ and it is calculated using $\mathbf{X}_{CSP} = \mathbf{W}^T \mathbf{X}$, where $\mathbf{W} \in \mathbb{R}^{n_{Chann} \times n_{Chann}}$ denotes the CSP matrix and n_{Chann} is the number of EEG channels that are used for

recording. The pre-processed EEG signals are denoted as $\mathbf{X} \in \mathbb{R}^{n_{Chann} \times n_{Samp}}$, where n_{Samp} is the number of sampled time points [18]. Since, in this work the EEG signals are collected from 16 channels, $\mathbf{W}$ is calculated from the 16 channels resulting in 16 CSP filters. We denote the matrix of the selected CSP filters (columns of $\mathbf{W}$) as $\mathbf{W}_{sel} | \mathbf{W}_{sel} \in \mathbb{R}^{n_{Chann} \times n_{FP}}$, where n_{FP} is the number of selected CSP filter pairs. The joint matrix $\mathbf{W}_{sel}$ is used to calculate the final CSP-filtered EEG signal using $\mathbf{X}_{CSP} = \mathbf{W}_{sel}^T \mathbf{X}$. For our case, $n_{FP} = 4$ gave the best accuracy, hence the first-and-last four column vectors of the $\mathbf{W}$ matrix are used.

E. Feature Extraction using FBCSP

FBCSP comprises of multiple band-pass filters that use zero-phase Chebyshev Type II filters and spatial filters that are used for CSP feature extraction [9]. Initially, a filter bank, with several sub-band filters, is employed to find the concerned EEG signals' features into these specific sub-bands. After that, each of these sub-bands are subjected to CSP feature extraction. Then, typically a feature selection method, here, mutual information-based feature selection (MIFS), is employed in order to select the most discriminative features which are generated by different filter banks. For this particular experiment, we have used five different frequency sub-bands, namely, [4–8]Hz, [8–12]Hz, [12–16]Hz, [16–32]Hz, [32–40]Hz.

F. Classification using Random Forest

Random Forest (RF) [25] is used as classifier to separate the MI tasks in our work. In RF, a predetermined number of Decision Trees (DT) are grown using randomly bootstrapped data. This means that for each DT there is a randomly selected subset of input features, which are considered with repetition. For each of this subsets there is a random number of input features that are considered for making the decision [26], [27]. Our RF has 475 DT and its hyper-parameters are tuned using grid search method. The RF classifier is trained using subject-wise 5-fold cross validation. The average classification accuracy is used as the performance metric.

IV. RESULTS & DISCUSSION

In this section the experimental results for our proposed method are shown for 6 stroke survivors. The experiments are performed to distinguish between the left and right wrist dorsiflexion MI tasks. The core idea behind this experiment is to compare the efficacy of the WPD with HOS method and the widely used CSP and FBCSP methods. Even though WPD has been previously used for MI-EEG signal classification, to the best of our knowledge it has never been used for stroke survivors MI-EEG data classification. The classification accuracies of the three different methods are shown in Table III. For all 6 subjects the WPD+HOS method performs better than both the CSP and FBCSP based method. The biggest improvement can be seen for Subject #1 and #6. The average accuracy achieved for WPD+HOS method is 71%, whereas for the CSP and FBCSP based method, the average accuracy

TABLE III
PERFORMANCE (ACCURACY IN %) COMPARISON FOR EACH INDIVIDUAL SUBJECT WHILE USING WPD, CSP AND FBCSP

	S1	S2	S3	S4	S5	S6	Avg. Acc.
WPD+HOS	73	71	68	72	70	72	71
CSP	64	70	63	67	69	62	66
FBCSP	67	71	64	70	71	68	69

achieved is 66% and 69%. A five-fold cross validation is used to select every event for testing at least once. Although for subject S2 and S5, FBCSP produces either same or better results than WPD+HOS based method, in terms of average accuracy FBCSP produces lower accuracy than the WPD+HOS based method. The variability in classification accuracy in each subject is due to varying levels of disability among the stroke survivors. The effect of stroke in their brain is different in each subject. Nonetheless, an average accuracy of more than 70% shows the efficiency of WPD+HOS based feature extraction technique over the standard CSP and FBCSP based methods. In comparison with the existing state-of-the-art methods, as discussed in Section II, our proposed method uses only 16 number of EEG channels' data which is much lower than most of these methods. Therefore, achieving an accuracy of more than 70% with 5-fold cross validation on 6 stroke subjects' data shows the efficacy of our proposed method.

V. CONCLUSION

This paper proposes a MI-EEG signal classification method based on WPD and HOS for stroke survivors. Commonly used feature extraction methods, the CSP and FBCSP based methods, are also used and compared here to show that WPD+HOS based method performs better. For all three types of feature extraction methods, the classification algorithm remains same with the use of RF. The experiments are performed on 6 stroke survivors' imaginary left and right wrist dorsiflexion EEG data. Experimental results show that the WPD+HOS based feature extraction method is able to achieve 71% average accuracy which is significantly higher than the CSP or FBCSP based methods. Therefore, it can be stated that, in future, the WPD+HOS based feature extraction method coupled with RF based classification can be considered for building a BCI assisted rehabilitation system for stroke survivors.

REFERENCES

- [1] A. Galimanis, M.-L. Mono, M. Arnold, K. Nedeltchev, and H. P. Mattle, "Lifestyle and stroke risk: a review," *Current Opinion in Neurology*, vol. 22, no. 1, p. 60–68, 2009.
- [2] C. O. Johnson, M. Nguyen, G. A. Roth, E. Nichols, T. Alam, D. Abate, F. Abd-Allah, A. Abdelalim, H. N. Abraha, N. M. Abu-Rmeileh, and et al., "Global, regional, and national burden of stroke, 1990–2016: a systematic analysis for the global burden of disease study 2016," *The Lancet Neurology*, vol. 18, no. 5, p. 439–458, 2019.
- [3] S. J. Page, P. Levine, S. Sisto, and M. V. Johnston, "A randomized efficacy and feasibility study of imagery in acute stroke," *Clinical Rehabilitation*, vol. 15, no. 3, p. 233–240, 2001.
- [4] M. A. Khan, R. Das, H. K. Iversen, and S. Puthusserypady, "Review on motor imagery based bci systems for upper limb post-stroke neurorehabilitation: From designing to application," *Computers in Biology and Medicine*, vol. 123, p. 103843, 2020.

- [5] T. C. Machado, A. A. Carregosa, M. S. Santos, N. M. da Silva Ribeiro, and A. Melo, "Efficacy of motor imagery additional to motor-based therapy in the recovery of motor function of the upper limb in post-stroke individuals: a systematic review," *Topics in Stroke Rehabilitation*, vol. 26, no. 7, pp. 548–553, 2019, pMID: 31264520. [Online]. Available: <https://doi.org/10.1080/10749357.2019.1627716>
- [6] S. H. Johnson, G. Sprehn, and A. J. Saykin, "Intact Motor Imagery in Chronic Upper Limb Hemiplegics: Evidence for Activity-Independent Action Representations," *Journal of Cognitive Neuroscience*, vol. 14, no. 6, pp. 841–852, 08 2002. [Online]. Available: <https://doi.org/10.1162/0899290202760191072>
- [7] T. Mulder, "Motor imagery and action observation: cognitive tools for rehabilitation," *Journal of Neural Transmission*, vol. 114, no. 10, p. 1265–1278, 2007.
- [8] N. Sharma, V. M. Pomeroy, and J.-C. Baron, "Motor imagery," *Stroke*, vol. 37, no. 7, p. 1941–1952, 2006.
- [9] R. Das, P. S. Lopez, M. Ahmed Khan, H. K. Iversen, and S. Puthusserypady, "FBCSP and adaptive boosting for multiclass motor imagery bci data classification: A machine learning approach," *2020 IEEE International Conference on Systems, Man, and Cybernetics (SMC)*, 2020.
- [10] D. Irimia, N. Sabathiel, R. Ortner, M. Poboroniuc, W. Coon, B. Z. Allison, and C. Guger, "recoverix: A new bci-based technology for persons with stroke," in *2016 38th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*, 2016, pp. 1504–1507.
- [11] C. Tangwiriyasakul, V. Mocioiu, M. J. van Putten, and W. L. Rutten, "Classification of motor imagery performance in acute stroke," *Journal of Neural Engineering*, vol. 11, no. 3, p. 036001, 2014.
- [12] Y. Liu, H. Zhang, M. Chen, and L. Zhang, "A boosting-based spatial-spectral model for stroke patients' eeg analysis in rehabilitation training," *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, vol. 24, no. 1, p. 169–179, 2016.
- [13] D. C. Irimia, R. Ortner, M. S. Poboroniuc, B. E. Ignat, and C. Guger, "High classification accuracy of a motor imagery based brain-computer interface for stroke rehabilitation training," *Frontiers in Robotics and AI*, vol. 5, p. 130, 2018. [Online]. Available: <https://www.frontiersin.org/article/10.3389/frobt.2018.00130>
- [14] V. K. Benzy, A. P. Vinod, R. Subasree, S. Alladi, and K. Raghavendra, "Motor imagery hand movement direction decoding using brain computer interface to aid stroke recovery and rehabilitation," *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, vol. 28, no. 12, p. 3051–3062, 2020.
- [15] P. Gaur, H. Gupta, A. Chowdhury, K. McCreddie, R. B. Pachori, and H. Wang, "A sliding window common spatial pattern for enhancing motor imagery classification in eeg-bci," *IEEE Transactions on Instrumentation and Measurement*, p. 1–1, 2021.
- [16] G. Feng, L. Hao, and G. Nuo, "Feature extraction algorithm based on csp and wavelet packet for motor imagery eeg signals," *2019 IEEE 4th International Conference on Signal and Image Processing (ICSIP)*, 2019.
- [17] C. Fan, X. Gu, X. Liu, and X. Li, "Motor imagery recognition method based on multi-classifier integrated decision," *2020 5th International Conference on Mechanical, Control and Computer Engineering (ICM-CCE)*, 2020.
- [18] X. Deng, B. Zhang, K. Liu, J. Wang, P. Yang, and C. Hu, "The classification of motor imagery eeg signals based on the time-frequency-spatial feature," *2021 IEEE 10th Data Driven Control and Learning Systems Conference (DDCLS)*, 2021.
- [19] A. Chowdhury, H. Raza, Y. K. Meena, A. Dutta, and G. Prasad, "On-line covariate shift detection-based adaptive brain-computer interface to trigger hand exoskeleton feedback for neuro-rehabilitation," *IEEE Transactions on Cognitive and Developmental Systems*, vol. 10, no. 4, pp. 1070–1080, 2018.
- [20] S. Aggarwal and N. Chugh, "Signal processing techniques for motor imagery brain computer interface: A review," *Array*, vol. 1-2, p. 100003, 2019.
- [21] J. Kevric and A. Subasi, "Comparison of signal decomposition methods in classification of eeg signals for motor-imagery bci system," *Biomedical Signal Processing and Control*, vol. 31, p. 398–406, 2017.
- [22] Y. Kutlu and D. Kuntalp, "Feature extraction for eeg heartbeats using higher order statistics of wpd coefficients," *Computer Methods and Programs in Biomedicine*, vol. 105, no. 3, p. 257–267, 2012.
- [23] B. Blankertz, F. Losch, M. Krauledat, G. Dornhege, G. Curio, and K.-R. Müller, "The berlin brain-computer interface: Accurate performance from first-session in bci-naïve subjects," *IEEE Transactions on Biomedical Engineering*, vol. 55, no. 10, pp. 2452–2462, 2008.
- [24] C. Sannelli, T. Dickhaus, S. Halder, E.-M. Hammer, K.-R. Müller, and B. Blankertz, "On optimal channel configurations for smr-based brain-computer interfaces," *Brain Topography*, vol. 23, no. 2, p. 186–193, 2010.
- [25] L. Breiman, "Random forests," *Machine Learning*, vol. 45, pp. 5–32, 2001.
- [26] M. Bentlemlsan, E.-T. Zemouri, D. Bouchaffra, B. Yahya-Zoubir, and K. Ferroudji, "Random forest and filter bank common spatial patterns for eeg-based motor imagery classification," in *2014 5th International Conference on Intelligent Systems, Modelling and Simulation*, 2014, pp. 235–238.
- [27] F. Li, Y. Fan, X. Zhang, C. Wang, F. Hu, W. Jia, and H. Hui, "Multi-feature fusion method based on eeg signal and its application in stroke classification," *Journal of Medical Systems*, vol. 44, no. 2, 2019.